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(TECHNICAL DETAILS)

**JOB NO: OT/NC/16/CIF (R)/6921/2026-27 : GENR SET 15 KVA (05 NOS)
IN THE AOR OF CIF (R)**

S. No	Brief Description
1.	Genr Set 15 KVA : Supply, Installing, Testing and Commissioning of 'Silent Type' Diesel Generating set having Prime Power Rating of 15 KVA, 415 volts at 1500 RPM, 0.8 lagging power factor at 415 V suitable for 50 Hz, 3 phase system, for 0.85 Load Factor with AMF Panel alongwith Acoustic Enclosure. Make : Ashok Leyland/Cummins/Genesis/Greaves Power/Kirloskar or equivalent to IS-10001-1981 (Reaffirmed-2002).
	(a) Diesel Engine. Diesel Engine shall be as per the following specifications :- (i) Rate Engine power (kWm) : 100% of the required powered at STP (Standard Temperature Pressure) ie equal to (Nominated Rate Capacity(KVA) of power generator + any Auxilary power Consumption by the Power generator) x Power factor (0.8)/Alternator efficiency. (ii) Type of Engine Cooling : Liquid Cooled. (iii) Number of cylinders (Nos) : 2 (iv) Number of Strokes (Nos) : 4 (v) Rated RPM of Engine (RPM) : 1500 (vi) Fuel : High Speed Diesel (HSD) (vii) Overload capacity : Engine is capable of delivering an output of 10% in excess of rated KVA for a period of one hour in any period of 12 hours continuous running. (viii) Salient Features of Engine : Direct injection Fuel System. Make : Ashok Leyland/Cummins/Genesis/Greaves Power/Kirloskar or equivalent to BS-649, BS-5514, IS-10001(1981) & amendment No-6 of Nov 2010.
	(b) Governing Class. Governing class shall be as per the following specifications :- (i) Type of governor : Mechanical (ii) Class of governor : A2 or better
	(c) Alternator. Alternator shall be as per the following specifications :- (i) Alternator Voltage Rating : 415 Volt (ii) Rating of AC Generator (KVA) : 15 (iii) Power Factor of AC Generator : 0.8 (iv) Conformity to Indian Standard (for Alternator) : Conforming to IS-13364 (Part-I) latest (upto 20 KVA) (v) Type of Alternator : Brushless (vi) Voltage Regulation Grade : VG3 (vii) IP Rating of Control Panel : IP 23 (viii) Class of Insulation : H Make : Kirloskar/Stamford/Crompton Greaves/Leroy Somer or equivalent to IS-4722/BS-2613/BS-5541/ISO-3046 : 2020
	(d) Control Panel. Control Panel shall be as per the following specifications :- (i) Control Panel : AMF Panel (ii) Control Panel Location : Inside the canopy (iii) IP Rating of Control Panel : IP 41 or more (iv) Display meters in the control panel (with appropriate rating and accuracy class) : Multifunctional Digital Display Meter (Displaying Voltage, Current, Frequency, Power Factor) (vii) Indicators : Low Fuel Level or more features (viii) Audio Alarm : Low Fuel Level or more features. Note :- "Reasonability of rate shall be responsibility of order placing authority"
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	(e) Acoustic Enclosure. Acoustic Enclosure shall be as per the following specifications :- (i) Sheet Thickness (mm) : 1.26 mm or more (ii) Thickness of insulation : 25 mm (Minimum) or more (iii) Density of insulation (kg/cubic m) : 20 or more (iv) Noise level at 1 meter (dB) : 75 (v) Salient Features : Glass window on Acoustic Enclosure in front of the Control Panel shall be provided.																
	(f) Fuel Tank. Fuel Tank shall be as per the following specifications :- (i) Fuel Tank Capacity : As per OEM specifications (ii) Number of Fuel Tank : 1 Nos (iii) Fuel Tank Sheet Material Thickness : 2 mm (Minimum) (iv) Fuel Tank Fabricated Material : MS Sheet (v) Fuel Tank Features : Fuel Tank capacity is sufficient for 8 hrs of running at full load.																
	(g) Battery. Battery shall be as per the following specifications :- (i) Battery Type & Specification : Low Maintenance free to IS-14247 for high cranking performance (ii) No of Batteries : 1 (iii) Capacity of Battery : As per OEM specifications																
	(h) Warranty/Services. Warranty/Services shall be as per the following specifications :- (i) Warranty on completion power generator/DG Set : 24 Months (ii) Warranty in running hours : 5000 hours (iii) Response Time to attend the complaint during warranty : 7 days																
	(j) Starting system :- 12V DC starting system comprising of starter motors: voltage regulator and arrangement for initial excitation complete with 1 Nos of batteries as required as per specifications.																
	(k) PCC Platform :- Genr Set shall be mounted on suitable base frame made of MS channel with Anti-vibration mountings arrangement with Genr Set placed on PCC platform of size 2.60m x 1.60m. The PCC platform shall be 150mm thick PCC (1:3:6) type C-1 using 20 mm graded stone agg over sub base of 150mm thick PCC (1:4:8) type D-2 using 40 mm graded stone agg over 150mm thick hardcore on rammed earth.																
2.	Earthing with each 15 KVA Genr set.. One earthing set shall be provided as per IS-3043-1987 for Each Genr Set as per nomenclature given below :- <table><tr><td>(i) Finished GI earthing plate 600 x 600 x 6 mm</td><td>- 01 No.</td></tr><tr><td>(ii) CI cover with frame 300 x 300 x 6mm thick</td><td>- 01 No.</td></tr><tr><td>(iii) GI Pipe 20 mm dia light</td><td>- 02 RM</td></tr><tr><td>(iv) GI strip 25 x 4 mm</td><td>- 08 RM</td></tr><tr><td>(v) Charcoal Dust</td><td>- 50 KG</td></tr><tr><td>(vi) Common Salt</td><td>- 50 KG</td></tr><tr><td>(vii) Nuts, Bolts and washer Size No- 12</td><td>- 08 Nos</td></tr><tr><td>(viii) GI funnel with GI wire mesh</td><td>- 01 No.</td></tr></table>	(i) Finished GI earthing plate 600 x 600 x 6 mm	- 01 No.	(ii) CI cover with frame 300 x 300 x 6mm thick	- 01 No.	(iii) GI Pipe 20 mm dia light	- 02 RM	(iv) GI strip 25 x 4 mm	- 08 RM	(v) Charcoal Dust	- 50 KG	(vi) Common Salt	- 50 KG	(vii) Nuts, Bolts and washer Size No- 12	- 08 Nos	(viii) GI funnel with GI wire mesh	- 01 No.
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3.	Two Years Maint Spares with each 15 KVA Genr Set (i) Fan Belt - 02 Nos (ii) Delivery Valve - 06 Nos (iii) High Pressure Pipe - 04 Nos (iv) Head Gasket - 02 Nos (v) Fuel Flexible Pipe - 06 Nos (vi) Benjo Bolt - 04 Nos (vii) Benjo Washer - 06 Nos (viii) Fuel Filter - 08 Nos (ix) Lub Oil Filter - 08 Nos (x) Lubrication Oil - 27 Ltrs (xi) Coolant - 7.5 Ltrs
4.	Tool Kit. Standard set of Tool Kit shall be provided for each Genr Set as per nomenclature given below :- (i) Plier side cutting 200mm - 01 Nos (ii) Screw driver 200mm - 01 Nos (iii) Screw driver 50mm - 01 Nos (iv) Spanner 10-11 - 01 Nos (v) Spanner 13-17 - 01 Nos (vi) Spanner 19-20 - 01 Nos (vii) Spanner 24-27 - 01 Nos (viii) Hammer 200gm - 01 Nos (ix) Adjustable Spanner 150mm - 01 Nos
5.	Set of accessories with each 15 KVA Genr Set. Set of accessories shall be provided for each Genr Set as per nomenclature given below :- (i) PVC insulated XLPE armoured cable 16 Sqmm, - 10 Mtr 3.5 core aluminium conductor multi strand. Make : HAVELLS/FINLOLEX/POLYCAB/ANCHOR/PLAZA or equivalent to IS-1554-1998. (ii) Main Switch 64 Amps, 415 V, TPN - 01 Nos Make : L&T/ABB/SIEMENS/ SCHNIENDER/LEGRAND/ C&S/HAVELLS or equivalent to IS - 13947-part-III-1993. (iii) Change Over Switch 64 Amps, 415 V, TPN - 01 Nos Make : L&T/ABB/SIEMENS/ SCHNIENDER/LEGRAND/ C&S/HAVELLS or equivalent to IS - 13947-part-III-1993.
6.	Construction Mtrl. (i) Cement. Cement of 43 grade OPC in sealed bags each weighting 50 Kgs. The date of manufacture of cement should not be more than one month prior to the date of delivery. Make: Ultra Tech/ Ambuja/ACC/JK Lakshmi or Equivalent to IS-8112-1989 (ii) Sand. River sand conforming to IS-2116-1980. (iii) Stone Agg 20mm. Stone aggregate conforming to IS 383-1970. (iv) Stone Agg 40mm. Stone aggregate conforming to IS 383-1970. (v) Hardcore. 40mm to 63mm broken boulder stone conforming to IS-383-1970.

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STORE/RATE LIST

JOB NO OT/NC/16/CIF (R)/6921/2026-27 : PROVISION OF 05 X GENR SET 15 KVA
IN THE AOR OF CIF (R)

Ser No	Items Title	Items	A/U	Qty
1	Genr Set 15 KVA Supply Installing Testing and Commissioning of Silent Type Diesel Generating set having Prime Power Rating of 15 KVA	Genr Set 15 KVA : Supply, Installing, Testing and Commissioning of ‘Silent Type’ Diesel Generating set having Prime Power Rating of 15 KVA, 415 volts at 1500 RPM, 0.8 lagging power factor at 415 V suitable for 50 Hz, 3 phase system, for 0.85 Load Factor with AMF Panel alongwith Acoustic Enclosure. Make : Ashok Leyland/Cummins/Genesis/Greaves Power/Kirloskar or equivalent to IS-10001-1981 (Reaffirmed-2002). Diesel Engine. Diesel Engine shall be as per the following specifications :- (i) Rate Engine power (kWm) : 100% of the required powered at STP (Standard Temperature Pressure) ie equal to (Nominated Rate Capacity(KVA) of power generator + any Auxilary power Consumption by the Power generator) x Power factor (0.8)/Alternator efficiency. (ii) Type of Engine Cooling : Liquid Cooled. (iii) Number of cylinders (Nos) : 2 (iv) Number of Strokes (Nos) : 4 (v) Rated RPM of Engine (RPM) : 1500 (vi) Fuel : High Speed Diesel (HSD) (vii) Overload capacity : Engine is capable of delivering an output of 10% in excess of rated KVA for a period of one hour in any period of 12 hours continuous running. (viii) Salient Features of Engine : Direct injection Fuel System. Make : Ashok Leyland/Cummins/Genesis/Greaves Power/ Kirloskar or equivalent to BS-649, BS-5514, IS-10001(1981) & amendment. Governing Class. Governing class shall be as per the following specifications :- (i) Type of governor : Mechanical (ii) Class of governor : A2 or better	Set	5

<u>Alternator.</u> Alternator shall be as per the following specifications :- (i) Alternator Voltage Rating : 415 Volt (ii) Rating of AC Generator (KVA) : 15 (iii) Power Factor of AC Generator : 0.8 (iv) Conformity to Indian Standard (for Alternator) : Conforming to IS-13364 (Part-I) latest (upto 20 KVA) (v) Type of Alternator : Brushless (vi) Voltage Regulation Grade : VG3 (vii) IP Rating of Control Panel : IP 23 (viii) Class of Insulation : H <u>Make</u> : Kirloskar/Stamford/Crompton Greaves/ Leroy Somer or equivalent to IS-4722/BS-2613/ BS-5541/ISO-3046(2020)
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		<u>Warranty/Services.</u> Warranty/Services shall be as per the following specifications :- (i) Warranty on completion power generator/DG Set : 24 Months (ii) Warranty in running hours : 5000 hours (iii) Response Time to attend the complaint during warranty : 7 days		
		<u>Starting system</u> :- 12V DC starting system comprising of starter motors: voltage regulator and arrangement for initial excitation complete with 1 Nos of batteries as required as per specifications.		
		<u>PCC Platform</u> :- Genr Set shall be mounted on suitable base frame made of MS channel with Anti-vibration mountings arrangement with Genr Set placed on PCC platform of size 2.60m x 1.60m. The PCC platform shall be 150mm thick PCC (1:3:6) type C-1 using 20 mm graded stone agg over sub base of 150mm thick PCC (1:4:8) type D-2 using 40 mm graded stone agg over 150mm thick hardcore on rammed earth.		
2	Finished GI earthing plate 600 x 600 x 6 mm	Earthing with each 15 KVA Genr set. One earthing set shall be provided as per IS-3043-1987 for Each Genr Set as per nomenclature given below :- (i) Finished GI earthing plate 600 x 600 x 6 mm - 01 No	Nos	5
3	CI cover with frame 300 x 300 x 6mm thick	(ii) CI cover with frame 300 x 300 x 6mm thick - 01 No	Nos	5
4	GI Pipe 20 mm dia light	(iii) GI Pipe 20 mm dia light - 02 RM	RM	10
5	GI strip 25 x 4 mm	(iv) GI strip 25 x 4 mm - 08 RM	RM	40
6	Charcoal Dust	(v) Charcoal Dust - 50 Kg	Kg	250
7	Common Salt	(vi) Common Salt - 50 Kg	Kg	250
8	Nuts Bolts and washer Size No 12	(vii) Nuts, Bolts and washer Size No-12 : 8 Nos	Nos	40
9	GI funnel with GI wire mesh	(viii) GI funnel with GI wire mesh - 01 No	Nos	5
10	Fan Belt	Two Years Maint Spares with each 15 KVA Genr Set (i) Fan Belt - 02 Nos	Nos	10
11	Delivery Valve	(ii) Delivery Valve - 06 Nos	Nos	30
12	High Pressure Pipe	(iii) High Pressure Pipe - 04 Nos	Nos	20
13	Head Gasket	(iv) Head Gasket - 02 Nos	Nos	10
14	Fuel Flexible Pipe	(v) Fuel Flexible Pipe - 06 Nos	Nos	30
15	Benjo Bolt	(vi) Benjo Bolt - 04 Nos	Nos	20
16	Benjo Washer	(vii) Benjo Washer - 06 Nos	Nos	30
17	Fuel Filter	(viii) Fuel Filter - 08 Nos	Nos	40
18	Lub Oil Filter	(ix) Lub Oil Filter - 08 Nos	Nos	40
19	Lubrication Oil	(x) Lubrication Oil - 27 Ltrs	Ltr	135
20	Coolant	(xi) Coolant - 7.5 Ltrs	Ltr	38
21	Plier side cutting 200mm	Tool Kit. Standard set of Tool Kit shall be provided for each Genr Set as per nomenclature given below :- (i) Plier side cutting 200mm	Nos	5
22	Screw driver 200mm	(ii) Screw driver 200mm	Nos	5
23	Screw driver 50mm	(iii) Screw driver 50mm	Nos	5
24	Spanner 10 11	(iv) Spanner 10-11	Nos	5

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25	Spanner 13 17	(v) Spanner 13-17	Nos	5
26	Spanner 19 20	(vi) Spanner 19-20	Nos	5
27	Spanner 24 27	(vii) Spanner 24-27	Nos	5
28	Hammer 200gm	(viii) Hammer 200gm	Nos	5
29	Adjustable Spanner 150mm	(ix) Adjustable Spanner 150mm	Nos	5
30	PVC insulated XLPE armoured cable 16 Sqmm, 3.5 core aluminium conductor multi strand	Set of accessories with each 15 KVA Genr Set. Set of accessories shall be provided for each Genr Set as per nomenclature given below :- (i) PVC insulated XLPE armoured cable 16 Sqmm, 3.5 core aluminium conductor multi strand. Make : HAVELLS/FINLOLEX/POLYCAB/ANCHOR/PLAZA or equivalent to IS-1554-1998.	RM	50
31	Main Switch 63 Amps, 415 V, TPN	(ii) Main Switch 63 Amps, 415 V, TPN Make : L&T/ABB/SIEMENS/ SCHNIENDER/ LEGRAND/ C&S/HAVELLS or equivalent to IS - 13947-part-III-1993.	Nos	5
32	Change Over Switch 63 Amps, 415 V, TPN	(iii) Change Over Switch 63 Amps, 415 V, TPN Make : L&T/ABB/SIEMENS/ SCHNIENDER/ LEGRAND/ C&S/HAVELLS or equivalent to IS - 13947-part-III-1993.	Nos	5

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